

# Non-Uniqueness of Distribution-Shift Attribution, and Why We Localize Subsets Instead

Technical note

## Abstract

For a marketplace comparing an existing batch of data against a new batch, we show a basic *impossibility*: the *total* distribution shift is identifiable and order-invariant, but the *per-feature contribution* to it is not uniquely defined — it depends on the order in which features are considered. We demonstrate this empirically with random feature-addition paths under two metrics: Maximum Mean Discrepancy (MMD) for covariate shift  $P(X)$  and the pseudo-outcome /  $R$ -risk for concept drift  $P(Y | X)$ . The consequence is methodological: at any level of the entity hierarchy one should *localize a responsible subset* (well-posed) rather than attribute per-feature proportions (ill-posed). We give the construction, the empirical result, the resulting multi-level drill-down, a worked business example, and how such a system should be evaluated.

## 1 Setup

Orders (observations) carry raw attributes and are labelled by batch  $W \in \{0, 1\}$  (existing vs. new). We roll orders up to an entity axis  $\mathbf{e}$  (merchant, buyer, item, region, ...) with a feature map  $\Phi$  producing an entity–feature matrix  $x \in \mathbb{R}^p$  (rich + rolling aggregations), with features grouped by raw attribute,  $g : \{1, \dots, p\} \rightarrow \mathcal{G}$ . Let  $P_0, P_1$  be the two-batch feature distributions. A characteristic-kernel MMD measures covariate shift,  $\text{MMD}^2(P_0, P_1) = 0 \iff P_0 = P_1$ ; with batch as “treatment”, the  $R$ -risk of the induced pseudo-outcome regression measures concept drift and is insensitive to pure covariate shift.

## 2 Path-order attribution and its non-uniqueness

Let  $T(S)$  be a shift metric (MMD<sup>2</sup> or  $R$ -risk) evaluated on the feature subset  $S \subseteq \{1, \dots, p\}$ . Given an ordering  $\pi$  of the features, the *addition path* is the curve  $S_0 = \emptyset \subset S_1 \subset \dots \subset S_p$  with  $S_k = \{\pi_1, \dots, \pi_k\}$ , and the marginal credited to feature  $\pi_k$  is  $T(S_k) - T(S_{k-1})$ . Sampling many random  $\pi$  and averaging the credits yields the Shapley value; but the credits along any single path are what a “how much did feature  $f$  contribute” question actually asks.

**Proposition 1** (Total identifiable, path non-unique). *The endpoint  $T(S_p) = T(\{1, \dots, p\})$  is invariant to the order  $\pi$ : the total shift is uniquely defined and estimable. The intermediate values  $T(S_k)$ ,  $0 < k < p$ , and hence the per-feature marginals, are not invariant to  $\pi$  whenever the features are dependent; there is no order-free “contribution of feature  $f$ ”. Shapley restores uniqueness only by averaging over all orders, i.e. by adopting a convention, not by identifying a truth.*

**Empirical demonstration.** We draw random feature-addition paths on the repository’s banded-Toeplitz covariate-shift and concept-drift DGMs and record the metric at every step. Figure 1 shows the result: all paths converge to the same full-set total (red, order-invariant), but the accrual curves

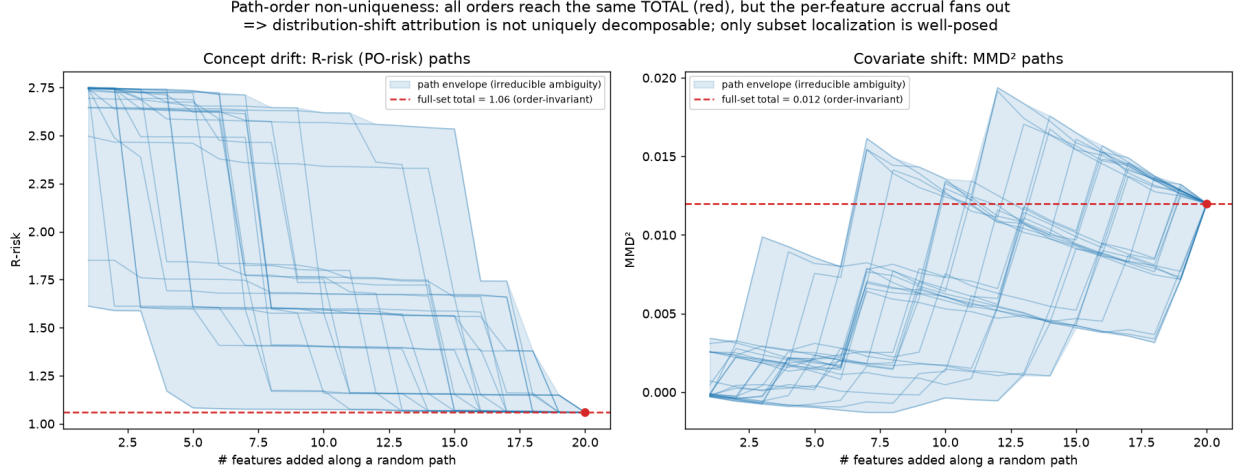


Figure 1: Random feature-addition paths. Every order reaches the same total (red, order-invariant); the per-feature accrual fans out (shaded envelope = irreducible ambiguity). Left: concept drift ( $R$ -risk); right: covariate shift ( $MMD^2$ ).

fan out in between; the shaded band is the *irreducible attribution ambiguity*. For concept drift the maximum gap between two orders is  $\approx$  the entire total ( $\sim 1.0$  against a total of  $\sim 1.06$ ): two feature orderings can disagree about “how much has been explained so far” by essentially the whole shift. The per-step standard deviation across orders is 30–40% of the total for both metrics.

**Remark 1** (Any number of levels). *The argument is scale-free: it applies to features within one entity, and equally to the coarser “how much does each child level contribute” question. Hence a distribution shift is not uniquely decomposable at any level of the hierarchy.*

### 3 The well-posed object: multi-level subset localization

While “how much” is not identifiable, “which subset carries the shift” is: the total is order-invariant, and the selection of the responsible group is stable across attribution conventions and across resamples. We therefore localize recursively, and never report proportions. The selection deliberately uses a single global detection plus a leave-one-group-out ablation and a stability check — *no multiple-testing machinery*, which keeps the procedure simple and easy to trust:

1. **Detect** the shift globally with one MMD /  $R$ -risk permutation test (order-invariant total).
2. **Localize (level 1)** to the responsible attribute-group subset by *leave-one-group-out (LOGO) ablation* — the groups whose removal collapses the shift — kept only if stable across bootstrap resamples.
3. **Drill (level 2)** by *restricting the order table to the entities selected at level 1* (a conditional drill-down: keep only the flagged subset’s rows), then re-running the same selection on the next aggregation key (e.g. item / product vertical) *within that subset* — again returning a subset, not a share. Level 2 is therefore always evaluated on the level-1 subset, not the whole population.

4. **Close the loop:** each selected subset carries an owner, a candidate rule (a KS-optimal split threshold), a decision (monitor / reweight / retrain / collect labels), and a trend status computed by diffing against the previous run.

## 4 Why LOGO, why these modalities, and why the subset is the unit

**Why LOGO.** We rank and select groups by the leave-one-group-out ablation  $\phi_g = T(\text{all}) - T(\text{all} \setminus g)$ , the drop in the shift metric when a group’s features are removed. It is the right primitive here for three reasons. (i) It answers the operational question directly — “if we ignore or repair this group, does the detected shift disappear?” — which is how an owner reasons and what makes the output actionable. (ii) It is a *conditional* importance (given all other features present), so a group is credited only for what the rest does not already explain: it flags *non-redundant* drivers rather than everything merely correlated with the shift. (iii) It is cheap ( $K+1$  metric evaluations), its sign is meaningful (removing a true driver lowers the metric; removing noise does not), and it needs no attribution convention. Applied at the raw-attribute *group* level it is immune to the within-group correlation that would otherwise dilute every single aggregation to near zero.

**Why the subset — not the feature — is the effective unit.** By the non-uniqueness result the total is order-invariant but the per-feature split is not; the object that *is* invariant and estimable is the contribution of a whole subset (include or exclude the group as a unit). Empirically the group-level answer is also stable across attribution conventions (standalone, LOGO, and Shapley agree on *which* groups) and across resamples, while the per-feature answer is not. The grouping is not arbitrary: a real shift typically enters through a raw signal — a collection/definition change in GMV, a pricing change, a new buyer cohort — which moves *all* of that signal’s aggregations together, so the raw-attribute group is the natural causal and operational unit. Selecting that unit is therefore both statistically well-posed (invariant, stable) and operationally meaningful (it maps to an owner, a rule, and an action).

**Why these modalities (MMD and PO-/R-risk).** A characteristic-kernel MMD is zero iff the two batches share the same  $P(X)$ , and its LOGO drop is a genuine reduction in a distributional distance that is sensitive to shifts in any moment (mean, scale, tails), so it does not miss non-linear covariate shifts. The PO-/R-risk targets the orthogonal, outcome-relevant question — has  $P(Y | X)$  changed — and is insensitive to pure covariate shift, so the two modalities cover the two failure modes without double-counting. Both are permutation-exact and assumption-light, which is what makes a LOGO drop built on them a trustworthy, convention-free number — in contrast to Shapley, which becomes unique only by averaging over an arbitrary set of orders.

## 5 A worked business example

**Symptom.** In the new week’s batch, the global test fires. Level-1 localization on the *region* axis selects the activity subset — concretely the order-count feature `n_orders` (and order-value aggregations) shifted *up* for a subset of users in region *R*: those buyers are placing more (and larger) orders than the existing batch.

**Non-uniqueness, honestly.** We do *not* claim “region  $R$ ’s users account for 43% of the shift” — by the proposition that number is order/convention dependent. We claim the shift is *localized to region  $R$ ’s user subset*, with a stability of, say, 0.98 and a split rule `n_orders per user > t`.

**Drill one level — *inside the flagged user subset*.** We first restrict the order table to exactly the level-1 subset (the flagged region- $R$  users, `n_orders per user > t`), and only then aggregate those users’ orders on the *item / product-vertical* axis and re-run the two-stage selection. This conditional drill localizes the extra orders to a specific **product vertical** (item category  $C$ ) *within that user subset*: among the flagged region- $R$  users the surge concentrates in category  $C$  (e.g. “fresh grocery”), not uniformly across the catalogue. The localization is thus **product-vertical**  $\times$  **user-subset** (category  $C$  conditioned on the region- $R$  user segment), output as `subset = {region  $R$  users, category  $C$ }` with its own split rule and stability. Evaluating level 2 on the whole population instead of on the level-1 subset would confound the two levels and is explicitly avoided.

**Action / closed loop.** The subset routes to the category- $C$  / region- $R$  owner; the candidate rule becomes a monitor; the decision (e.g. *reweight* if the mapping is unchanged, *collect labels / retrain* if the outcome relationship moved) is attached; the **trend** field marks it *resolved* once the subset stops flagging. The business learns *where* and *who* and *what to do* — never a spurious percentage.

**Remark 2** (Why this is the right granularity). “43% from category  $C$ ” would be *unfalsifiable (non-identifiable) and un-actionable*. “The new-batch order surge is localized to category  $C$  among region- $R$  users; here is the threshold and the owner” is *checkable, routable, and monitorable* — it closes the loop.

**Results of the two-level procedure.** We inject a GMV covariate shift concentrated in a *single product item category* (Fresh Grocery) for new merchants (200 merchants, 6 categories, 24k orders) and run the procedure. Level 1 (merchant axis) selects the **gmV** attribute group by LOGO ablation with full stability; the conditional Level 2 then drills into the **product item category** and isolates **Fresh Grocery** exactly, while every other category is non-significant (Table 1). No proportions are reported — only the selected subsets, their ablation/gap magnitude, stability, and permutation  $p$ .

## 6 How to evaluate such a system

Because per-feature proportions are not identifiable, they must not be the yard stick. Evaluate only identifiable targets:

| Target                 | Identifiable?           | Metric                                     |
|------------------------|-------------------------|--------------------------------------------|
| Total / detection      | yes (order-invariant)   | test power; type-I under permuted labels   |
| Subset localization    | yes (convention-stable) | subset recall / precision vs. ground truth |
| Stability              | yes                     | bootstrap selection frequency              |
| Decision layer         | yes (with labels)       | downstream $\Delta$ loss backtest          |
| Per-feature proportion | <b>no</b>               | — (do not evaluate)                        |

The principle: judge a method by the questions it claims to answer (detection, subset localization, stability, downstream impact), not by a question that is provably unanswerable. The non-identifiability itself is demonstrated empirically (Figure 1), not hidden.

| Level (unit)                                                                  | Candidate                                | LOGO / gap   | Stability | Perm. $p$    |
|-------------------------------------------------------------------------------|------------------------------------------|--------------|-----------|--------------|
| <i>Level 1 — merchant axis</i> (global MMD <sup>2</sup> = 0.281, $p$ = 0.003) |                                          |              |           |              |
|                                                                               | <b>gmv</b> (selected)                    | +0.283       | 1.00      | —            |
|                                                                               | quantity                                 | −0.277       | 0.85      | —            |
| <i>Level 2 — product item category</i> (conditional on gmv, new vs. existing) |                                          |              |           |              |
|                                                                               | <b>Fresh Grocery</b> ( $c^*$ , selected) | <b>55.88</b> | —         | <b>0.002</b> |
|                                                                               | Home & Kitchen                           | 0.77         | —         | 0.158        |
|                                                                               | Apparel                                  | 0.58         | —         | 0.240        |
|                                                                               | Electronics                              | 0.29         | —         | 0.621        |
|                                                                               | Beauty                                   | 0.26         | —         | 0.583        |
|                                                                               | Toys                                     | 0.03         | —         | 0.932        |

Table 1: Two-level conditional subset localization (`fsds_cross_level.py`). Level 1 selects the **gmv** attribute subset (LOGO ablation + bootstrap stability); the conditional Level 2 drills into the **product item category** and selects exactly the shifted one, **Fresh Grocery** (gap = new−existing mean GMV within the category; merchant-block permutation  $p$ ). The order surge is localized to a single product item category, not spread as proportions; ground truth is recovered at both levels.

## 7 Conclusion

The total distribution shift is identifiable; its per-feature decomposition is not. This is not a limitation of a particular estimator but a property of dependent features under any additive path attribution. The productive response is to *localize responsible subsets* recursively down the entity hierarchy and attach owner / rule / decision / trend to each — turning an unanswerable “how much” into an operational “which subset, who, and what to do”.

To summarize, this **feature selection for distribution shift (FSDS)** procedure can be extended beyond two batches of data, unlocking the potential for multi-layer feature selection and localization in both feature stores and concurrent metric systems. In many circumstances, the vague associations between metrics are not yielded by business resource-intensive models (which require a lot of man-days from the business requirement to the deployment) and are not deployed in production systems — this is where another value of our FSDS procedure lies.